Tools UniProtKB Advanced | List Search Help

# P08581 · MET_HUMAN

**Protein**[i]
**Hepatocyte growth factor receptor**

**Gene**[i]
**MET**

**Status**[i]
UniProtKB reviewed (Swiss-Prot)

**Organism**[i]
Homo sapiens (Human)

**Amino acids**
1390 (go to sequence)

**Protein existence**[i]
Evidence at protein level

**Annotation score**[i]
5/5

Tools Download Add Add a publication Entry feedback

## Function[i]

Receptor tyrosine kinase that transduces signals from the extracellular matrix into the cytoplasm by binding to hepatocyte growth factor/HGF ligand. Regulates many physiological processes including proliferation, scattering, morphogenesis and survival. Ligand binding at the cell surface induces autophosphorylation of MET on its intracellular domain that provides docking sites for downstream signaling molecules. Following activation by ligand, interacts with the PI3-kinase subunit PIK3R1, PLCG1, SRC, GRB2, STAT3 or the adapter GAB1. Recruitment of these downstream effectors by MET leads to the activation of several signaling cascades including the RAS-ERK, PI3 kinase-AKT, or PLCgamma-PKC. The RAS-ERK activation is associated with the morphogenetic effects while PI3K/AKT coordinates prosurvival effects. During embryonic development, MET signaling plays a role in gastrulation, development and migration of neuronal precursors, angiogenesis and kidney formation. During skeletal muscle development, it is crucial for the migration of muscle progenitor cells and for the proliferation of secondary myoblasts (By similarity).
In adults, participates in wound healing as well as organ regeneration and tissue remodeling. Also promotes differentiation and proliferation of hematopoietic cells. May regulate cortical bone osteogenesis (By similarity). By Similarity

(Microbial infection) Acts as a receptor for Listeria monocytogenes internalin InlB, mediating entry of the pathogen into cells. 1 Publication 2 Publications

### Catalytic activity[i]

**Rhea:10596**

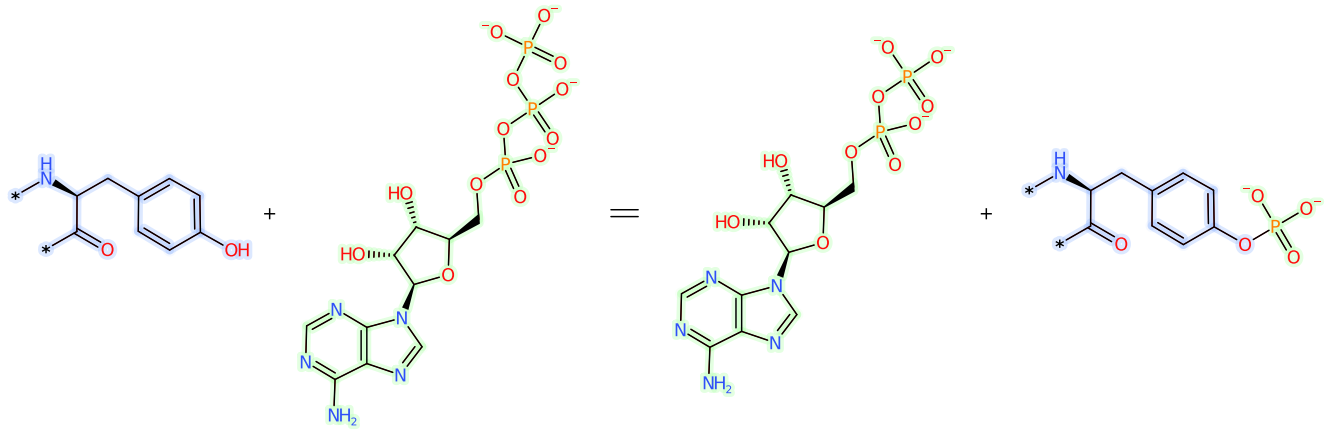
L-tyrosyl-[protein] + ATP = O-phospho-L-tyrosyl-[protein] + ADP + $H^+$ PROSITE-ProRule Annotation

EC:2.7.10.1 (UniProtKB | ENZYME | Rhea)

Hide reaction details

Reaction | Atom map

Color atoms ☑, Number atoms ☐, Confidence = 0.28048



This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our Privacy Notice.

I agree, dismiss this banner

## Activity regulation[i]

In its inactive state, the C-terminal tail interacts with the catalytic domain and inhibits the kinase activity. Upon ligand binding, the C-terminal tail is displaced and becomes phosphorylated, thus increasing the kinase activity.

## Features

Showing features for site[i], binding site[i], active site[i].

| | | | | |
|---|---|---|---|---|
| + | Site | 307-308 | Cleavage Sequence Analysis | |
| + | Site | 1003 | Required for ligand-induced CBL-mediated ubiquitination 1 Publication | |
| + | Site | 1009-1010 | Breakpoint for translocation to form TPR-MET oncogene | |
| + | Binding site | 1084-1092 | ATP (UniProtKB \| ChEBI ) PROSITE-ProRule Annotation | Tools Add |
| + | Binding site | 1110 | ATP (UniProtKB \| ChEBI ) | |
| + | Active site | 1204 | Proton acceptor PROSITE-ProRule Annotation | |

## Gene Ontology[i]

GO annotations | GO-CAM models New

TYPE

± All

ID POSITION(S) DESCRIPTION

Animals and fungi (Opisthokonta)

all annotations | All Molecular Function | DNA binding | DNA-binding transcription factor activity | RNA binding | catalytic activity | signaling receptor binding | structural molecule activity | transporter activity | cytoskeletal protein binding | transcription factor binding | lipid binding | enzyme regulator activity | carbohydrate binding | small molecule binding | signaling receptor activity | metal ion binding | carbohydrate derivative binding | Other Molecular Function | All Biological Process | immune system process | carbohydrate metabolic process | DNA metabolic process | lipid metabolic process | cell cycle | behavior | cell population proliferation | catabolic process | programmed cell death | cellular component organization | RNA metabolic process | protein metabolic process | reproductive process | signaling | cell differentiation | developmental process | homeostatic process | nervous system process | response to stimulus | establishment of localization | carbohydrate derivative metabolic process | Other Biological Process

Cell color indicative of number of GO terms

| ASPECT | TERM |
|---|---|
| Molecular Function | ATP binding Source:UniProtKB-KW |
| Molecular Function | hepatocyte growth factor receptor activity Source:GO_Central |
| Molecular Function | identical protein binding Source:IntAct 2 Publications |
| Molecular Function | molecular function activator activity Source:DisProt 1 Publication |
| Molecular Function | protein phosphatase binding Source:UniProtKB 1 Publication |
| Molecular Function | protein tyrosine kinase activity Source:Reactome |
| Molecular Function | semaphorin receptor activity Source:InterPro |
| Biological Process | branching morphogenesis of an epithelial tube Source:UniProtKB 1 Publication |
| Biological Process | cell surface receptor protein tyrosine kinase signaling pathway Source:GO_Central |
| Biological Process | cell surface receptor signaling pathway Source:UniProtKB 1 Publication |
| Biological Process | endothelial cell morphogenesis Source:UniProtKB 1 Publication |
| Biological Process | excitatory postsynaptic potential Source:BHF-UCL |

Expand table

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our Privacy Notice.

PAN-GO
P08581 14 GO annotations based on evolutionary models

## Keywords[i]

**Molecular function**
#Kinase
#Receptor
#Transferase
#Tyrosine-protein kinase

**Ligand**
#ATP-binding
#Nucleotide-binding

## Enzyme and pathway databases

BRENDA
2.7.10.1 2681

PathwayCommons
P08581

Reactome
R-HSA-1257604 PIP3 activates AKT signaling
R-HSA-2219530 Constitutive Signaling by Aberrant PI3K in Cancer
R-HSA-416550 Sema4D mediated inhibition of cell attachment and migration
R-HSA-5673001 RAF/MAP kinase cascade
R-HSA-6806942 MET Receptor Activation
More Reactome links

SIGNOR
P08581

SignaLink
P08581

ENZYME
Search...

## Protein family/group databases

TCDB
8.A.23.1.45 the basigin (basigin) family

# Names & Taxonomy[i]

## Protein names[i]

**Recommended name**
**Hepatocyte growth factor receptor**

**EC number**
EC:2.7.10.1 (UniProtKB | ENZYME | Rhea )

**Short name**
HGF receptor

**Alternative names**
HGF/SF receptor
Proto-oncogene c-Met
Scatter factor receptor (SF receptor)
Tyrosine-protein kinase Met

## Gene names[i]

**Name**
MET

## Organism names

**Taxonomic identifier[i]**
9606 (NCBI )

**Organism[i]**
Homo sapiens (Human)

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our Privacy Notice.

Eukaryota > Metazoa > Chordata > Craniata > Vertebrata > Euteleostomi > Mammalia > Eutheria > Euarchontoglires > Primates > Haplorrhini > Catarrhini > Hominidae > Homo

## Accessions[i]

**Primary accession**
P08581

**Secondary accessions**
A1L467
B5A932
E7EQ94
O60366
Q12875
More accessions

## Proteomes[i]

**Identifier**
UP000005640

**Component[i]**
Chromosome 7

## Organism-specific databases

**AGR**
HGNC:7029

**HGNC**
HGNC:7029 MET

**MIM**
114550 phenotype
164860 gene
605074 phenotype
607278 phenotype
616705 phenotype
620019 phenotype

**VEuPathDB**
HostDB:ENSG00000105976

# Subcellular Location[i]

UniProt Annotation | GO Annotation

Membrane ; Single-pass type I membrane protein

## Isoform 3

Secreted

## Features

Showing features for topological domain[i], transmembrane[i].

| | | | | |
|---|---|---|---|---|
| + | Topological domain | 25-932 | Extracellular Sequence Analysis | Tools Add |
| + | Transmembrane | 933-955 | Helical Sequence Analysis | Tools Add |
| + | Topological domain | 956-1390 | Cytoplasmic Sequence Analysis | Tools Add |

## Keywords[i]

TYPE

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our Privacy Notice.

# Disease & Variants[i]

## Involvement in disease[i]

Activation of MET after rearrangement with the TPR gene produces an oncogenic protein

Defects in MET may be associated with gastric cancer

### Hepatocellular carcinoma (HCC)

1 Publication

**Note**

The disease is caused by variants affecting the gene represented in this entry

**Description**

A primary malignant neoplasm of epithelial liver cells. The major risk factors for HCC are chronic hepatitis B virus (HBV) infection, chronic hepatitis C virus (HCV) infection, prolonged dietary aflatoxin exposure, alcoholic cirrhosis, and cirrhosis due to other causes.

**See also**

MIM:114550

**Natural variants in HCC**

| | | | | |
|---|---|---|---|---|
| VAR_032490 | 1173 | T>I | in HCC; dbSNP:rs121913675 | 1 Publication |
| VAR_032492 | 1244 | K>R | in HCC; dbSNP:rs121913677 | 1 Publication |
| VAR_032493 | 1250 | M>I | in HCC; dbSNP:rs121913676 | 1 Publication |

### Renal cell carcinoma papillary (RCCP)

5 Publications

| VARIANT ID | POSITION(S) | CHANGE | DESCRIPTION |
|---|---|---|---|

**Note**

The disease is caused by variants affecting the gene represented in this entry

**Description**

A subtype of renal cell carcinoma tending to show a tubulo-papillary architecture formed by numerous, irregular, finger-like projections of connective tissue. Renal cell carcinoma is a heterogeneous group of sporadic or hereditary carcinoma derived from cells of the proximal renal tubular epithelium.

**See also**

MIM:605074

**Natural variants in RCCP**

| VARIANT ID | POSITION(S) | CHANGE | DESCRIPTION |
|---|---|---|---|
| VAR_032485 | 1092 | V>I | in RCCP; constitutive autophosphorylation; dbSNP:rs786202724 3 Publications |
| VAR_032486 | 1094 | H>L | in RCCP; constitutive autophosphorylation; causes malignant transformation in cell lines; dbSNP:rs121913243 1 Publication |
| VAR_032487 | 1094 | H>R | in RCCP; causes malignant transformation in cell lines; dbSNP:rs121913243 2 Publications |
| VAR_032488 | 1094 | H>Y | in RCCP; constitutive autophosphorylation; causes malignant transformation in cell lines; dbSNP:rs121913244 1 Publication |
| VAR_032489 | 1106 | H>D | in RCCP; constitutive autophosphorylation; causes malignant transformation in cell lines; dbSNP:rs1795030662 2 Publications |
| VAR_006286 | 1131 | M>T | in RCCP; germline mutation; dbSNP:rs121913668 2 Publications |
| VAR_006287 | 1188 | V>L | in RCCP; germline mutation; dbSNP:rs121913669 2 Publications |
| VAR_006288 | 1195 | L>V | in RCCP; somatic mutation; dbSNP:rs121913673 1 Publication |
| VAR_006289 | 1220 | V>I | in RCCP; germline mutation; dbSNP:rs121913670 1 Publication |

Expand table

A common allele in the promoter region of the MET shows genetic association with susceptibility to autism in some families. Functional assays indicate a decrease in MET promoter activity and altered binding of specific transcription factor complexes

MET activating mutations may be involved in the development of a highly malignant, metastatic syndrome known as cancer of unknown primary origin (CUP) or primary occult malignancy. Systemic neoplastic spread is generally a late event in cancer progression. However, in some instances, distant dissemination arises at a very

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

1 Publication

**Note**

The disease is caused by variants affecting the gene represented in this entry

**Description**

A form of non-syndromic sensorineural hearing loss with prelingual onset. Sensorineural deafness results from damage to the neural receptors of the inner ear, the nerve pathways to the brain, or the area of the brain that receives sound information.

**See also**

MIM:616705

**Natural variants in DFNB97**

| VARIANT ID | POSITION(S) | CHANGE | DESCRIPTION |
|---|---|---|---|
| VAR_075757 | 841 | F>V | in DFNB97; dbSNP:rs794728016 1 Publication |

### Osteofibrous dysplasia (OSFD)

1 Publication

**Note**

Disease susceptibility is associated with variants affecting the gene represented in this entry. Disease-associated variants identified in 4 families cause the deletion of exon 14. This results in the exclusion of an ubiquitination target site within the cytoplasmic domain, hence in protein stabilization. The persistent presence of MET at the cell surface in conditions of ligand-dependent activation retards osteoblastic differentiation 1 Publication

**Description**

A congenital disorder of osteogenesis characterized by non-neoplastic, radiolucent lesions that affect the cortical bone immediately under the periosteum. It usually manifests as a painless swelling or anterior bowing of the long bones, most commonly the tibia and fibula.

**See also**

MIM:607278

**Natural variants in OSFD**

| VARIANT ID | POSITION(S) | CHANGE | DESCRIPTION |
|---|---|---|---|
| VAR_076584 | 964-1010 | *missing* | in OSFD; loss of CBL-mediated destabilization 1 Publication |

### Arthrogryposis, distal, 11 (DA11)

1 Publication

**Note**

The disease may be caused by variants affecting the gene represented in this entry

**Description**

A form of distal arthrogryposis, a disease characterized by congenital joint contractures that mainly involve two or more distal parts of the limbs, in the absence of a primary neurological or muscle disease. DA11 is an autosomal dominant form characterized mainly by camptodactyly. Other features include absent flexion creases and limited forearm supination.

**See also**

MIM:620019

**Natural variants in DA11**

| VARIANT ID | POSITION(S) | CHANGE | DESCRIPTION |
|---|---|---|---|
| VAR_087543 | 1234 | Y>C | in DA11; not phosphorylated in response to HGF; severely decreased tyrosin kinase activity; dbSNP:rs1554400286 1 Publication |

Showing features for natural variant[i], mutagenesis[i].

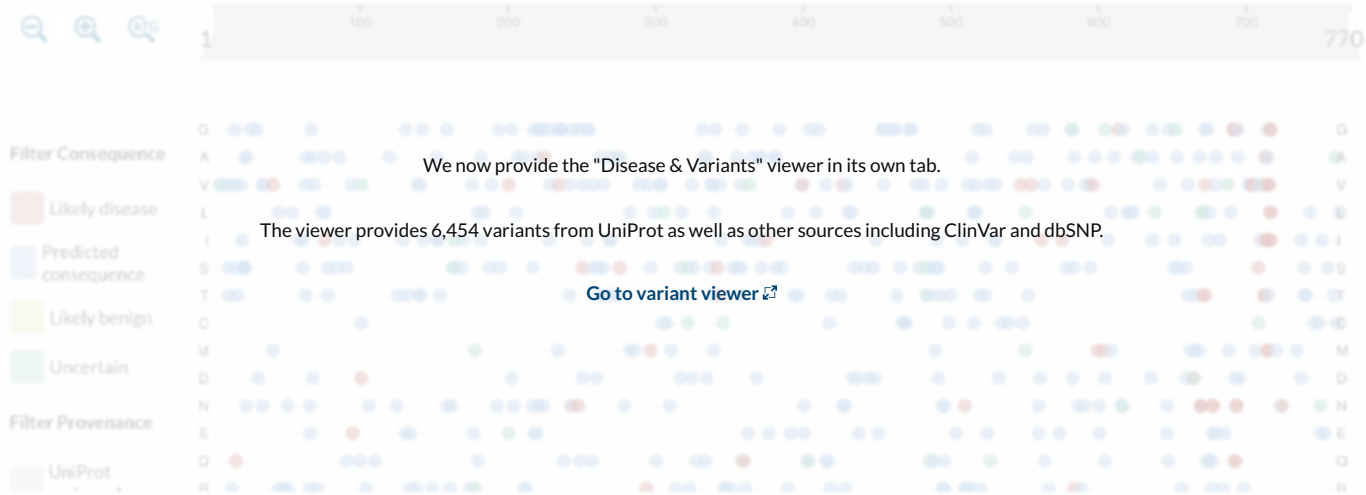
| ± ▾ | TYPE All | ID | POSITION(S) | DESCRIPTION |
|---|---|---|---|---|
| + | Natural variant | VAR_041738 | 143 | in dbSNP:rs35469582 1 Publication |
| + | Natural variant | VAR_064855 | 150 | found in a case of cancer of unknown primary origin; uncertain significance; |

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our Privacy Notice.

| | TYPE | ID | POSITION(S) | DESCRIPTION |
|---|---|---|---|---|
| + | Natural variant | VAR_032478 | 238 | in dbSNP:rs34349517 |
| + | Natural variant | VAR_032479 | 316 | in dbSNP:rs35225896 |

Expand table

## Variants




We now provide the "Disease & Variants" viewer in its own tab.

The viewer provides 6,454 variants from UniProt as well as other sources including ClinVar and dbSNP.

Go to variant viewer

## Keywords[i]

**Disease**
#Deafness
#Disease variant
#Non-syndromic deafness
#Proto-oncogene

## Organism-specific databases

**DisGeNET**
4233

**MIM**
114550 phenotype
164860 gene
605074 phenotype
607278 phenotype
616705 phenotype
620019 phenotype

**MalaCards**
MET

**OpenTargets**
ENSG00000105976

**Orphanet**
47044 Hereditary papillary renal cell carcinoma
488265 Osteofibrous dysplasia
319298 Papillary renal cell carcinoma
33402 Pediatric hepatocellular carcinoma
90636 Rare autosomal recessive non-syndromic sensorineural deafness type DFNB

## Miscellaneous

**Pharos**
P08581 Tclin

## Chemistry

**ChEMBL**
CHEMBL3717

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

**DrugBank**
**DB06896** 1-(4-fluorophenyl)-N-[3-fluoro-4-(1H-pyrrolo[2,3-b]pyridin-4-yloxy)phenyl]-2-oxo-1,2-dihydropyridine-3-carboxamide
**DB08791** 1-[(2-NITROPHENYL)SULFONYL]-1H-PYRROLO[3,2-B]PYRIDINE-6-CARBOXAMIDE
**DB06997** 2-(4-fluorophenyl)-N-{[3-fluoro-4-(1H-pyrrolo[2,3-b]pyridin-4-yloxy)phenyl]carbamoyl}acetamide
**DB07969** 3-[3-(4-methylpiperazin-1-yl)-7-(trifluoromethyl)quinoxalin-5-yl]phenol
**DB17191** Altiratinib
More DrugBank links

**DrugCentral**
**P08581**

**GuidetoPHARMACOLOGY**
**1815**

## Genetic variation databases

**BioMuta**
**MET**

**ClinGen**
**HGNC:7029**

**DMDM**
**251757497**

**GenCC**
**HGNC:7029**

# PTM/Processing[i]

## Features

Showing features for signal[i], chain[i], modified residue (large scale data)[i], glycosylation[i], disulfide bond[i], modified residue[i].

| ± | TYPE (All) | ID | POSITION(S) | SOURCE (All) | DESCRIPTION | |
|---|---|---|---|---|---|---|
| + | Signal | | 1-24 | UniProt | Sequence Analysis | Tools Add |
| + | Chain | PRO_0000024440 | 25-1390 | UniProt | Hepatocyte growth factor receptor | Tools Add |
| + | Modified residue (large scale data) | | 27 | PTMeXchange | Ubiquitinated lysine Bronze[i] Combined Sources | |
| + | Modified residue (large scale data) | | 32 | PTMeXchange | Ubiquitinated lysine Silver[i] Combined Sources | |
| + | Glycosylation | | 45 | UniProt | N-linked (GlcNAc...) asparagine Sequence Analysis | |

Expand table

## Post-translational modification[i]

Autophosphorylated in response to ligand binding on Tyr-1234 and Tyr-1235 in the kinase domain leading to further phosphorylation of Tyr-1349 and Tyr-1356 in the C-terminal multifunctional docking site. Dephosphorylated by PTPRJ at Tyr-1349 and Tyr-1365. Dephosphorylated by PTPN1 and PTPN2. 5 Publications

Ubiquitinated. Ubiquitination by CBL regulates MET endocytosis, resulting in decreasing plasma membrane receptor abundance, and in endosomal degradation and/or recycling of internalized receptors. 1 Publication 1 Publication

O-mannosylation of IPT/TIG domains by TMEM260 is required for protein maturation (PubMed:37186866).
O-mannosylated residues are composed of single mannose glycans that are not elongated or modified (PubMed:37186866). 1 Publication

(Microbial infection) Tyrosine phosphorylation is stimulated by L.monocytogenes InlB. Tyrosine phosphorylation is maximal 10-20 minutes after treatment with InlB and disappears by 60 minutes. The phosphorylated residues were not identified. 1 Publication

## Keywords[i]

**PTM**
**#Disulfide bond**
**#Glycoprotein**
**#Phosphoprotein**

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

**CPTAC**
CPTAC-1496
CPTAC-2784
CPTAC-2816

**MassIVE**
P08581

**PaxDb**
9606-ENSP00000317272

**PeptideAtlas**
P08581

**ProteomicsDB**
52128 [P08581-1]
52129 [P08581-2]
52130 [P08581-3]

**Pumba**
P08581

**jPOST**
P08581

### 2D gel databases

**OGP**
P08581

### PTM databases

**GlyConnect**
680 17 N-Linked glycans (6 sites)

**GlyCosmos**
P08581 13 sites, 23 glycans

**GlyGen**
P08581 21 sites, 72 N-linked glycans (10 sites), 2 O-linked glycans (3 sites)

**PhosphoSitePlus**
P08581

**SwissPalm**
P08581

**iPTMnet**
P08581

## Expression[i]

### Tissue specificity[i]

Expressed in normal hepatocytes as well as in epithelial cells lining the stomach, the small and the large intestine. Found also in basal keratinocytes of esophagus and skin. High levels are found in liver, gastrointestinal tract, thyroid and kidney. Also present in the brain. Expressed in metaphyseal bone (at protein level) (PubMed:26637977). 3 Publications

### Gene expression databases

**Bgee**
ENSG00000105976 Expressed in pigmented layer of retina and 193 other cell types or tissues

**ExpressionAtlas**
P08581 baseline and differential

### Organism-specific databases

**HPA**
ENSG00000105976 Tissue enhanced (liver)

## Interaction[i]

### Subunit[i]

Heterodimer made of an alpha chain (50 kDa) and a beta chain (145 kDa) which are disulfide linked. Binds PLXNB1. Interacts when phosphorylated with downstream effectors including STAT3, PIK3R1, SRC, PCLG1, GRB2 and GAB1. Interacts with SPSB1, SPSB2 and SPSB4 (By similarity).
Interacts with INPP5D/SHIP1. When phosphorylated at Tyr-1356, interacts with INPPL1/SHIP2. Interacts with RANBP9 and RANBP10, as well as SPSB1, SPSB2, SPSB3 and SPSB4. SPSB1 binding occurs in the presence and in the absence of HGF, however HGF treatment has a positive effect on this interaction. Interacts with MUC20; prevents interaction with GRB2 and suppresses hepatocyte growth factor-induced cell proliferation. Interacts with GRB10. Interacts with PTPN1 and PTPN2. Interacts with LECT2; this interaction may have an antagonistic effect on receptor activation (PubMed:27334921).
Interacts with HSP90AA1 and HSP90AB1; the interaction suppresses MET kinase activity (PubMed:26517842).
Interacts with tensin TNS3 (PubMed:24814316).
Interacts (when phosphorylated) with tensin TNS4 (via SH2 domain); the interaction increases MET protein stability by inhibiting MET endocytosis and subsequent lysosomal degradation (PubMed:24814316). By Similarity 19 Publications

(Microbial infection) Interacts via extracytoplasmic residues 25-656 with L.monocytogenes InlB; MET can bind HGF, its endogenous ligand, and InlB simultaneously (PubMed:11081636, PubMed:17662939).

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our Privacy Notice.

| TYPE All | ENTRY 1 | ENTRY 2 | NUMBER OF EXPERIMENTS | INTACT |
|---|---|---|---|---|
| BINARY | P08581 | CBL P22681 | 17 | EBI-1039152, EBI-518228 |
| BINARY | P08581 | CDH1 P12830 | 2 | EBI-1039152, EBI-727477 |
| BINARY | P08581 | CTNND1 O60716 | 2 | EBI-1039152, EBI-701927 |
| BINARY | P08581 | DNAJA3 Q96EY1 | 4 | EBI-1039152, EBI-356767 |
| BINARY | P08581 | DNAJA3 Q96EY1-2 | 2 | EBI-1039152, EBI-3952284 |
| BINARY | P08581 | EGFR P00533 | 8 | EBI-1039152, EBI-297353 |
| BINARY | P08581 | FGR P09769 | 2 | EBI-1039152, EBI-1383732 |
| BINARY | P08581 | HGF P14210 | 7 | EBI-1039152, EBI-1039104 |
| BINARY | P08581 | HGF P14210-6 | 3 | EBI-1039152, EBI-6280319 |
| XENO | P08581 | Hgf Q08048 | 3 | EBI-1039152, EBI-15655650 |

Expand table

## Protein-protein interaction databases

**BioGRID**
110391 279 interactors

**CORUM**
P08581

**DIP**
DIP-6023N

**ELM**
P08581

**FunCoup**
P08581 1057 interactors

**IntAct**
P08581 203 interactors

**MINT**
P08581

**STRING**
9606.ENSP00000317272

## Chemistry

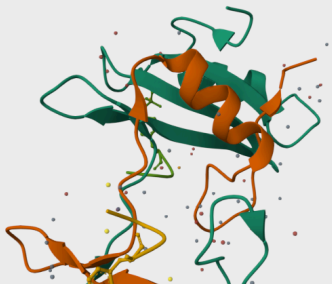
**BindingDB**
P08581

## Miscellaneous

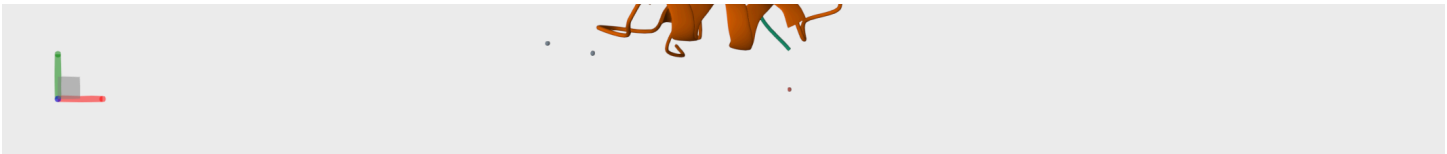
**RNAct**
P08581 protein

## Structure[i]

View UniProt features on this structure in the Feature Viewer.



This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our Privacy Notice.



| SOURCE -- Select -- | IDENTIFIER | ISOFORM | METHOD -- Select -- | RESOLUTION | CHAIN | POSITIONS | LINKS |
|---|---|---|---|---|---|---|---|
| PDB | 1FYR | | X-ray | 2.40 Å | I/J/K/L | 1356-1359 | PDBe · RCSB-PDB · PDBj · PDBsum . Foldseek |
| PDB | 1R0P | | X-ray | 1.80 Å | A | 1049-1360 | PDBe · RCSB-PDB · PDBj · PDBsum . Foldseek |
| PDB | 1R1W | | X-ray | 1.80 Å | A | 1049-1360 | PDBe · RCSB-PDB · PDBj · PDBsum . Foldseek |

## 3D structure databases

**AlphaFoldDB**
P08581

**EMDB**
EMD-23919
EMD-23920
EMD-23921
EMD-23922
EMD-23923

**SASBDB**
P08581

**SMR**
P08581

**ModBase**
Search...

**PDBe-KB**
Search...

## Miscellaneous

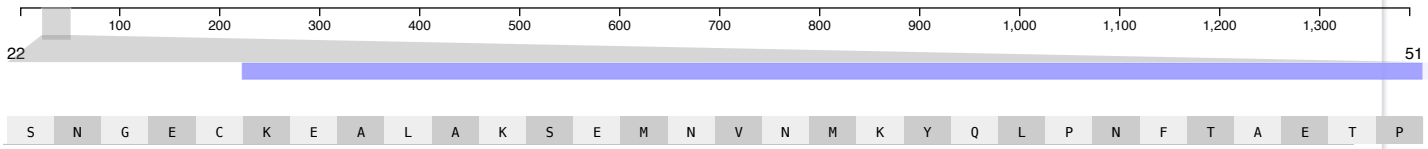
**EvolutionaryTrace**
P08581

# Family & Domains[i]

## Features

Showing features for domain[i], region[i].

Download




| | | | | |
|---|---|---|---|---|
| + | Domain | 27-515 | Sema PROSITE-ProRule Annotation | Tools Add |
| + | Domain | 563-655 | IPT/TIG 1 | Tools Add |
| + | Domain | 657-739 | IPT/TIG 2 | Tools Add |
| + | Domain | 742-836 | IPT/TIG 3 | Tools Add |
| + | Domain | 1078-1345 | Protein kinase PROSITE-ProRule Annotation | Tools Add |

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our Privacy Notice.

## Domain[i]

The kinase domain is involved in SPSB1 binding.

The beta-propeller Sema domain mediates binding to HGF.

## Sequence similarities[i]

Belongs to the **protein kinase superfamily**. **Tyr protein kinase family**. PROSITE-ProRule Annotation

## Keywords[i]

**Domain**

| ± ▾ | TYPE All | ID | POSITION(S) | DESCRIPTION |
|---|---|---|---|---|

# Transmembrane helix

## Family and domain databases

- View the Phylogenomic databases for this entry within the **Similar Proteins section**.
- View all family and domain features for this entry's canonical sequence in the **UniParc Feature Viewer**.

**CDD**
cd00603 IPT_PCSR 1 hit
cd01180 IPT_plexin_repeat1 1 hit
cd01179 IPT_plexin_repeat2 1 hit
cd01181 IPT_plexin_repeat3 1 hit
cd05058 PTKc_Met_Ron 1 hit
cd11278 Sema_MET 1 hit

**DisProt**
DP02738

**FunFam**
1.10.510.10:FF:000093 Hepatocyte growth factor receptor 1 hit
2.130.10.10:FF:000088 Hepatocyte growth factor receptor 1 hit
2.60.40.10:FF:000213 Hepatocyte growth factor receptor 1 hit
2.60.40.10:FF:000400 Hepatocyte growth factor receptor 1 hit
2.60.40.10:FF:002708 Hepatocyte growth factor receptor 1 hit
More FunFam links

**Gene3D**
2.60.40.10 Immunoglobulins 3 hits
3.30.200.20 Phosphorylase Kinase, domain 1 1 hit
1.10.510.10 Transferase(Phosphotransferase) domain 1 1 hit
2.130.10.10 YVTN repeat-like/Quinoprotein amine dehydrogenase 1 hit

**IDEAL**
IID00704

**InterPro**
View protein in InterPro
IPR013783 Ig-like_fold
IPR014756 Ig_E-set
IPR002909 IPT_dom
IPR011009 Kinase-like_dom_sf
More InterPro links

**PANTHER**
PTHR22625:SF61 HEPATOCYTE GROWTH FACTOR RECEPTOR 1 hit
PTHR22625 PLEXIN 1 hit

**PIRSF**
PIRSF000617 TyrPK_HGF-R 1 hit

**PRINTS**
PR00109 TYRKINASE

**PROSITE**
View protein in PROSITE
PS00107 PROTEIN_KINASE_ATP 1 hit
PS50011 PROTEIN_KINASE_DOM 1 hit
PS00109 PROTEIN_KINASE_TYR 1 hit
PS51004 SEMA 1 hit

**Pfam**
View protein in Pfam
PF07714 PK_Tyr_Ser-Thr 1 hit
PF01437 PSI 1 hit
PF01403 Sema 1 hit
PF01833 TIG 3 hits

**SMART**
View protein in SMART
SM00429 IPT 4 hits
SM00423 PSI 1 hit
SM00630 Sema 1 hit
SM00219 TyrKc 1 hit

**SUPFAM**
SSF81296 E set domains 3 hits
SSF103575 Plexin repeat 1 hit
SSF56112 Protein kinase-like (PK-like) 1 hit
SSF101912 Sema domain 1 hit

**MobiDB**
Search...

# Sequence & Isoforms[i]

Align isoforms (3) Add isoforms

**Sequence status**[i] **Sequence processing**[i]

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

## P08581-1

This isoform has been chosen as the **canonical** sequence. All positional information in this entry refers to it. This is also the sequence that appears in the downloadable versions of the entry.

**Name**
1

**See also**
sequence in **UniParc** or sequence clusters in **UniRef**

**Length**
1,390

**Mass (Da)**
155,541

**Last updated**
2009-07-07 v4

**MD5 Checksum**[i]
EE0753C4F68E188BDDB0C66890BEEE23

MKAPAVLAPGILVLLFTLVQRSNGECKEALAKSEMNVNMKYQLPNFTAETPIQNVILHEHHIFLGATNYIYVLNEEDLQKVAEYKTGPVLEHPDCFPCQDCSSKANLSGGVWK
DNINMALVVDTYYDDQLISCGSVNRGTCQRHVFPHNHTADIQSEVHCIFSPQIEEPSQCPDCVVSALGAKVLSSVKDRFINFFVGNTINSSYFPDHPLHSISVRRLKETKDGFM
FLTDQSYIDVLPEFRDSYPIKYVHAFESNNFIYFLTVQRETLDAQTFHTRIIRFCSINSGLHSYMEMPLECILTEKRKKRSTKKEVFNILQAAYVSKPGAQLARQIGASLNDDILFGV
FAQSKPDSAEPMDRSAMCAFPIKYVNDFFNKIVNKNNVRCLQHFYGPNHEHCFNRTLLRNSSGCEARRDEYRTEFTTALQRVDLFMGQFSEVLLTSISTFIKGDLTIANLGTSE
GRFMQVVVSRSGPSTPHVNFLLDSHPVSPEVIVEHTLNQNGYTLVITGKKITKIPLNGLGCRHFQSCSQCLSAPPFVQCGWCHDKCVRSEECLSGTWTQQICLPAIYKVFPNS
APLEGGTRLTICGWDFGFRRNNKFDLKKTRVLLGNESCTLTLSESTMNTLKCTVGPAMNKHFNMSIIISNGHGTTQYSTFSYVDPVITSISPKYGPMAGGTLLTLTGNYLNSGNS
RHISIGGKTCTLKSVSNSILECYTPAQTISTEFAVKLKIDLANRETSIFSYREDPIVYEIHPTKSFISGGSTITGVGKNLNSVSVPRMVINVHEAGRNFTVACQHRSNSEIICCTTPSLQ
QLNLQLPLKTKAFFMLDGILSKYFDLIYVHNPVFKPFEKPVMISMGNENVLEIKGNDIDPEAVKGEVLKVGNKSCENIHLHSEAVLCTVPNDLLKLNSELNIEWKQAISSTVLGK
VIVQPDQNFTGLIAGVVSISTALLLLLGFFLWLKKRKQIKDLGSELVRYDARVHTPHLDRLVSARSVSPTTEMVSNESVDYRATFPEDQFPNSSQNGSCRQVQYPLTDMSPILTSG
DSDISSPLLQNTVHIDLSALNPELVQAVQHVVIGPSSLIVHFNEVIGRGHFGCVYHGTLLDNDGKKIHCAVKSLNRITDIGEVSQFLTEGIIMKDFSHPNVLSLLGICLRSEGSPLV
VLPYMKHGDLRNFIRNETHNPTVKDLIGFGLQVAKGMKYLASKKFVHRDLAARNCMLDEKFTVKVADFGLARDMYDKEYYSVHNKTGAKLPVKWMALESLQTQKFTTKS
DVWSFGVLLWELMTRGAPPYPDVNTFDITVYLLQGRRLLQPEYCPDPLYEVMLKCWHPKAEMRPSFSELVSRISAIFSTFIGEHYVHVNATYVNVKCVAPYPSLLSSEDNADD
EVDTRPASFWETS

## P08581-2

**Name**
2

**See also**
sequence in **UniParc** or sequence clusters in **UniRef**

**Differences from canonical**
**755-755:** S → STWWKEPLNIVSFLFCFAS 1 Publication

## P08581-3

**Name**
3

**Synonyms**
Soluble MET variant 4

**See also**
sequence in **UniParc** or sequence clusters in **UniRef**

**Differences from canonical**
**755-764:** SGGSTITGVG → RHVNIALIQR 1 Publication
**765-1390:** Missing 1 Publication

## Computationally mapped potential isoform sequences[i]

There are 3 potential isoforms mapped to this entry

Align Add **View all**

| Entry | Entry name | Gene name | Length |
|---|---|---|---|
| C9JKM5 | C9JKM5_HUMAN | MET | 190 |
| E6Y365 | E6Y365_HUMAN | MET | 934 |
| H7C130 | H7C130_HUMAN | MET | 158 |

## Features

Showing features for sequence conflict[i], alternative sequence[i].

± ▾ TYPE All ID POSITION(S) DESCRIPTION

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

| | TYPE | ID | POSITION(S) | DESCRIPTION | |
|---|---|---|---|---|---|
| + | Sequence conflict | | 720 | in Ref. 3; ACF47606 Curated | |
| + | Alternative sequence | VSP_005005 | 755 | in isoform 2 1 Publication | |
| + | Alternative sequence | VSP_042447 | 755-764 | in isoform 3 1 Publication | Tools Add |
| + | Alternative sequence | VSP_042448 | 765-1390 | in isoform 3 1 Publication | Tools Add |
| + | Sequence conflict | | 1191 | in Ref. 1; AAA59591 Curated | |
| + | Sequence conflict | | 1272 | in Ref. 1; AAA59591, 2; CAB56793 and 6; AAA59590 Curated | |

## Keywords[i]

**Coding sequence diversity**
#Alternative splicing
#Chromosomal rearrangement

**Technical term**
#3D-structure
#Proteomics identification
#Reference proteome

## Sequence databases

**CCDS**
CCDS43636.1 [P08581-1]
CCDS47689.1 [P08581-2]

**PIR**
A40175 TVHUME

**RefSeq**
NP_000236.2 NM_000245.3 [P08581-1]
NP_001120972.1 NM_001127500.3 [P08581-2]

| NUCLEOTIDE SEQUENCE | PROTEIN SEQUENCE | MOLECULE TYPE | STATUS |
|---|---|---|---|
| J02958<br>EMBL · GenBank · DDBJ | AAA59591.1<br>EMBL · GenBank · DDBJ | mRNA | |
| X54559<br>EMBL · GenBank · DDBJ | CAB56793.1<br>EMBL · GenBank · DDBJ | mRNA | |
| EU826570<br>EMBL · GenBank · DDBJ | ACF47606.1<br>EMBL · GenBank · DDBJ | mRNA | |
| AC002080<br>EMBL · GenBank · DDBJ | AAB54047.1<br>EMBL · GenBank · DDBJ | Genomic DNA | |
| AC002543<br>EMBL · GenBank · DDBJ | AAC60383.1<br>EMBL · GenBank · DDBJ | Genomic DNA | |
| AC004416<br>EMBL · GenBank · DDBJ | AAF66137.2<br>EMBL · GenBank · DDBJ | Genomic DNA | |
| CH236947 | EAL24359.1 | Genomic DNA | |

Expand table

## Genome annotation databases

**Ensembl**
ENST00000318493.11 ENSP00000317272.6 ENSG00000105976.17 [P08581-2]
ENST00000397752.8 ENSP00000380860.3 ENSG00000105976.17 [P08581-1]
ENST00000436117.3 ENSP00000410980.2 ENSG00000105976.17 [P08581-3]

**GeneID**
4233

**KEGG**
hsa:4233

**MANE-Select**
ENST00000397752.8 ENSP00000380860.3 NM_000245.4 NP_000236.2

**UCSC**
uc003vij.4 human [P08581-1]

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our Privacy Notice.

## UniRef clusters[i]

## Orthologs & paralogs[i]

## Phylogenomic databases

**GeneTree**
ENSGT00940000158022

**HOGENOM**
CLU_005158_0_0_1

**InParanoid**
P08581

**OMA**
DEEPGQC

**OrthoDB**
9985181at2759

**PAN-GO**
P08581 14 GO annotations based on evolutionary models

**PhylomeDB**
P08581

**eggNOG**
KOG1095 Eukaryota
KOG3610 Eukaryota

## Organism-specific databases

**AGR**
HGNC:7029

## Disclaimer

*Any medical or genetic information present in this entry is provided for research, educational and informational purposes only. It is not in any way intended to be used as a substitute for professional medical advice, diagnosis, treatment or care. Our staff consists of biologists and biochemists that are not trained to give medical advice.*

UniProt
EMBL-EBI PIR SIB
Get in touch
UniProt is an ELIXIR core data resource
UniProt is a GBC global core biodata resource

| Core data | Supporting data | Tools | Information |
|---|---|---|---|
| Proteins (UniProtKB) | Literature citations | BLAST | Cite UniProt |
| Species (Proteomes) | Taxonomy | Align | About & Help |
| Protein clusters (UniRef) | Keywords | Retrieve/ID mapping | UniProtKB manual |
| Sequence archive (UniParc) | Subcellular locations | Peptide search | Technical corner |
| | Cross-referenced databases | Tool results | Expert biocuration |
| | Diseases | | Statistics |

Main funding by: National Institutes of Health EMBL-EBI State Secretariat for Education, Research and Innovation SERI

Release 2026_01 | Statistics
© 2002 – 2026 UniProt consortium
License & Disclaimer | Privacy Notice

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our Privacy Notice.